Summary

The study sense is a computing system designed to track environmental conditions and occupancy across multiple buildings. Each building ideally ideally is equipped with i dedicated Raspberry Pi unit featuring a camera and environmental sensors. The system performs local AI inference to count occupancy while gathering temperature, humidity, and noise levels. This telemetry data is published via MQTT to a cloud dashboard (ThingsBoard) and is simultaneously routed to a custom local website. The local site aggregates the MQTT data streams from all the individual Raspberry Pis, providing a unified, centralized view of the entire multi-building setup.

Hardware Architecture

The system is built on a decentralized, multi-node architecture, with an independent computing node deployed in each monitored building. Each node consists of:

- **Compute Core:** A Raspberry Pi handling multi-threaded Python execution, local AI inference, and MQTT network communication.
- **Vision Sensor:** Raspberry Pi Camera Module (accessed via Picamera2), capturing high-resolution (1440x1080) RGB frames for object detection.
- **Environmental Sensors:** GrovePi expansion board hosting a

DHT Sensor for Temperature and Humidity, and an Analog Sound Sensor for ambient noise level monitoring.

Software Stack & Technologies

- **YOLOv8 (Ultralytics):** AI object detection model used to analyze camera frames and count bounding boxes classified as persons.
- **OpenCV (cv2):** Computer Vision library used to manipulate image arrays and render the local live feed.
- **Paho MQTT:** Messaging protocol client used to publish JSON-formatted sensor telemetry to both the ThingsBoard cloud infrastructure and the local aggregated site.
- **Threading:** Python module allowing the camera feed and AI/Sensor processing to run concurrently without frame-blocking.

Data Routing: Cloud and Local Aggregation

A central feature of this architecture is its reporting mechanism using the MQTT protocol, which caters to both individual building tracking and campus-wide aggregation.

ThingsBoard Cloud Integration

Each Raspberry Pi connects securely to the ThingsBoard MQTT

broker. The background thread aggregates the occupancy, temperature, humidity, and noise level into a Python dictionary, serializes it into a JSON payload, and publishes it to the standard telemetry topic. This provides a remote, cloud-based dashboard for checking the health and stats of each individual device.

Local Site Aggregation (Multi-Building Hub)

In parallel to the cloud routing, the data is uploaded via MQTT to a custom-built local website. Because each building possesses its own standalone Raspberry Pi node, this local site serves as a centralized hub. It is configured to subscribe to the MQTT topics of all the distributed building nodes, aggregating their real-time data streams into a single, unified interface. This design allows administrators to monitor the real-time status of multiple library buildings simultaneously from one localized dashboard.

Technical Implementation: Concurrency & State Management

To ensure operational stability, the system relies on a multi-threaded architecture:

- **Main Thread (UI & Camera):** Runs continuously at a high frame rate, capturing arrays from the camera, drawing the OpenCV UI, and updating the local display monitor.
- **Background Thread (Sensors & AI):** Operates on a periodic

loop (e.g., every 10 seconds). It copies the latest camera frame, executes the YOLO model prediction, queries the physical GrovePi sensors, and pushes the consolidated payload to the MQTT brokers.

To prevent race conditions, where the main thread writes a new frame to the image array at the exact microsecond the background thread attempts to read it for YOLO inference, the codebase implements a strict thread lock. The background thread must acquire this lock before it can safely copy the image matrix into memory, thereby ensuring data integrity without freezing the video feed.